

BOUCTOUCHE, NB, Canada

WMO: 716660

Lat: 46.431N

Lon: 64.767W

Elev: 36

StdP: 100.89

Time Zone: -4.00 (NAA)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.5	-19.2	-26.4	0.3	-20.7	-23.9	0.4	-18.0	11.3	-3.0	10.2	-2.1	3.6	230	0.592

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	28.9	20.8	27.3	19.9	25.7	19.0	22.1	26.9	21.3	25.3	20.4	24.0	4.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.6	15.3	24.3	19.8	14.5	23.8	18.9	13.8	22.8	65.1	27.4	62.0	25.3	58.8	24.0	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.5	8.3	7.3	DB	-25.8	32.1	2.4	1.6	-27.5	33.2	-28.9	34.1	-30.2	35.0	-32.0	36.1	(4)
(5)			WB	-26.0	23.8	2.4	0.6	-27.7	24.2	-29.1	24.6	-30.4	25.0	-32.1	25.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.1	-7.4	-7.1	-2.8	3.8	10.2	15.4	19.9	19.0	14.7	8.5	2.5	-4.0	(6)	
	DBStd	10.61	6.44	5.78	5.13	4.29	4.14	3.53	3.02	2.87	4.01	4.14	5.04	5.68	(7)	
	HDD10.0	2406	539	479	397	193	48	2	0	0	6	78	229	433	(8)	
	HDD18.3	4595	798	713	655	436	253	102	18	28	121	304	474	692	(9)	
	CDD10.0	998	0	0	1	7	56	163	308	278	147	33	5	0	(10)	
	CDD18.3	143	0	0	0	0	2	13	68	47	13	1	0	0	(11)	
	CDH23.3	999	0	0	1	2	37	101	475	299	83	2	0	0	(12)	
	CDH26.7	199	0	0	0	0	6	15	111	52	16	0	0	0	(13)	
(14) Wind	WSAvg	3.5	3.9	3.9	4.1	3.9	3.7	3.2	2.9	2.8	3.2	3.4	3.8	3.8	(14)	
(15) Precipitation	PrecAvg	1128	89	80	93	82	88	93	97	86	101	109	96	105	(15)	
	PrecMax	1459	148	127	196	142	180	168	165	195	218	194	177	245	(16)	
	PrecMin	898	35	27	42	37	27	30	33	26	30	31	41	39	(17)	
	PrecStd	142	30	27	35	27	36	33	30	43	50	50	37	41	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.9	9.9	14.7	21.9	27.3	28.8	31.4	30.0	28.9	23.0	18.1	12.5	(19)	
		MCWB	10.9	7.9	8.5	14.9	17.5	18.7	22.2	21.5	21.8	17.5	16.6	11.4	(20)	
	2%	DB	7.1	5.8	8.9	16.8	23.7	26.1	29.3	28.2	25.6	19.5	15.3	8.7	(21)	
		MCWB	6.4	4.3	5.7	12.1	15.7	18.4	21.3	20.5	19.3	15.1	13.4	7.9	(22)	
	5%	DB	4.1	3.1	6.2	13.9	20.6	24.1	27.8	26.5	23.5	17.5	12.9	5.9	(23)	
		MCWB	3.0	1.3	3.6	10.1	14.0	17.8	20.5	19.6	18.6	14.8	10.8	5.0	(24)	
	10%	DB	1.4	1.0	4.1	11.2	17.8	22.3	26.1	24.9	21.4	15.4	10.3	3.6	(25)	
		MCWB	0.5	-0.1	1.8	7.3	12.6	16.5	19.5	18.8	17.5	12.7	8.4	2.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.0	8.5	10.5	15.9	19.3	21.4	23.5	23.0	22.1	18.5	16.8	11.6	(27)	
		MCDB	11.8	9.7	13.8	18.9	24.5	25.6	29.0	27.8	27.2	20.5	18.0	12.3	(28)	
	2%	WB	6.6	4.5	6.6	13.1	17.1	20.1	22.4	21.8	20.7	16.7	13.9	8.1	(29)	
		MCDB	7.1	5.8	8.3	15.4	21.6	24.0	27.7	25.9	23.8	18.7	15.0	8.7	(30)	
	5%	WB	3.0	1.8	3.9	10.4	15.3	18.8	21.6	20.9	19.4	14.9	11.2	5.2	(31)	
		MCDB	3.9	2.9	5.7	12.9	18.9	22.6	26.2	24.6	22.3	17.0	12.5	5.8	(32)	
	10%	WB	0.6	0.1	1.9	7.7	13.4	17.4	20.7	19.9	18.2	13.0	8.4	2.8	(33)	
		MCDB	1.3	0.8	4.0	10.7	17.2	21.0	24.5	23.5	21.2	14.8	9.8	3.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	9.4	8.8	9.0	10.4	9.7	10.0	10.1	10.4	9.0	7.9	7.6	(35)	
		MCDBR	11.7	10.7	11.0	14.5	15.8	12.9	12.2	12.4	12.2	11.4	11.1	9.7	(36)	
	5% WB	MCWBR	11.5	9.5	8.5	10.1	9.1	7.3	5.9	6.4	7.3	8.3	9.7	9.7	(37)	
		MCWBR	11.5	10.8	10.1	12.9	13.3	11.2	10.4	10.0	10.6	9.9	10.4	9.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.287	0.283	0.320	0.346	0.367	0.387	0.414	0.389	0.356	0.327	0.306	0.284	(40)	
		taud	2.412	2.388	2.389	2.418	2.408	2.345	2.307	2.371	2.451	2.530	2.535	2.498	(41)	
	Ebn at Noon		815	898	908	912	899	878	848	857	859	839	793	784	(42)	
		Edh at Noon	70	86	101	108	113	121	125	112	95	76	63	58	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.42	2.12	3.33	4.20	4.96	5.35	5.50	4.81	3.84	2.42	1.34	1.04	(44)	
	RadStd		0.13	0.21	0.21	0.42	0.52	0.41	0.38	0.32	0.34	0.24	0.16	0.10	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page